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A Real Network Environment Dataset for Traffic Analysis

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The objective of internet traffic analysis is to identify latent patterns and ascertain the true state of internet operations by examining traffic data. This approach is considered an effective and valuable means to achieve accurate network management. Whilst the extant network traffic datasets are predominantly collated within a laboratory environment, exhibiting deficiencies with regard to authenticity in terms of network scales, users, behaviours, and temporal and spatial characteristics, this paper proposes an *in-situ* network deployment and data collection scheme involving a large number of devices and users. The scheme involves the collection of a large real Internet traffic dataset including encrypted and non-encrypted traffic through sensors deployed on real-world network access equipment. Through desensitization, cleaning, feature engineering and labelling, an open database is created for researchers in the field of traffic analysis to use in academic and engineering.

Background & Summary

The Internet has evolved into an essential infrastructure for human life and production, driven by the integration of cloud networks, the expansion of network scale, and the proliferation of diverse network applications. Operator urban network traffic, growing annually by 15%[1,2], underscores the critical need to understand key information components and traffic patterns in cloud network traffic. This understanding is vital for enhancing operational value, improving data transformation capabilities, enabling automation and intelligent operations in cloud networks, and enhancing overall customer satisfaction. Concurrently, the rise in various network attack activities, including denial of service attacks, computer worms, and ransomware, poses significant cybersecurity risks[3–5].

The industry trend towards automated extraction of valuable service application information from existing network traffic reflects the growing demand for user traffic patterns and network security protection. Network traffic, as a data stream reflecting user network usage patterns, is readily accessible. However, the sheer volume of network data generated continuously on the Internet, such as the substantial data produced by short videos like TikTok every 10 minutes, necessitates a more efficient approach. Relying on individually developed methods and systems to collect network traffic data would be resource-intensive. Therefore, leveraging publicly available datasets that have been cleaned, extracted, and refined for service application analysis is imperative[6–11].

While mainstream public datasets like ISCXVPN016 and CSE-CIC-ISDS2018 offer valuable insights, they are static offline datasets collected in simulated test environments without real-time updates[1–7]. In contrast, our datasets[12] are derived from real-time Internet user interaction traffic data collected directly from the real network, rather than through PC or mobile device testing. After anonymizing key user information, our data processing and service application classification models, deployed online on plugins or platforms, extract original service application traffic data and corresponding labels based on session flow dimensions. The processed data is stored in PCAP or CSV formats and collected quasi real-time through terminal plugins integrated with server platforms[1,10,13].

For model training and general statistical data analysis, understanding the service application labels associated with each traffic session is crucial. Therefore, labeling traffic data by session and filtering out irrelevant session flows is necessary. Service application tags typically consist of two levels: the primary application category (e.g., web browsing, network video) and specific subcategories (e.g., TikTok, Douyu, iQiyi). Deep packet inspection (DPI) and deep flow inspection (DFI) technologies are commonly used in the industry to annotate network traffic data[8,9].

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Deep packet inspection and identification. DPI technology, known for its straightforward algorithm and quick deployment, boasts high application identification accuracy, reaching up to 95% for HTTP streams, making it the prevailing standard for traffic analysis. DPI technology primarily scrutinizes the application protocol of traffic by monitoring TCP and HTTP flows to extract essential field values, such as the host and user agent fields. These field values typically contain website URLs with the application name. In engineering, DPI technology is commonly implemented through two methods: mirroring technology, which duplicates data from access devices to servers for unpacking and analysis, and spectroscopic technology, which replicates data through spectral analysis, dividing it between access devices and CR before copying it to servers for further analysis[7–10].

Despite its advantages, DPI technology has several drawbacks. Firstly, the substantial amount of data collected and stored in full spectrum or mirror mode includes vast user service data with limited analytical value, leading to high storage and computing costs. DPI technology's reliance on flow tracking necessitates a significant server infrastructure to support analytical and computational tasks, resulting in substantial investments. Secondly, extracting keyword information from non-encrypted traffic requires analyzing all user data, potentially compromising user privacy, including account information, and introducing security risks. Lastly, with approximately 60% of real network traffic being encrypted, DPI technology struggles to identify encrypted traffic, such as tunneling protocols like HTTPS and anonymous network services that conceal critical information like application signatures.

Deep flow inspection and identification. DFI technology, an emerging traffic analysis approach, differs from DPI by employing a model training and learning methodology[13–15]. Instead of establishing fixed traffic identification rules based on data, DFI technology extracts features from original traffic data, processes these features, and utilizes them as input for training learning models. Subsequently, unknown traffic data is fed into the trained model for prediction and identification. The effectiveness of DFI technology hinges on the analysis and processing of traffic characteristics, which influences model selection and prediction performance.

DFI technology employs two primary feature extraction methods for network traffic identification: feature statistics-based methods and sequence-based methods. The feature statistics-based approach extracts features from statistical characteristics of each packet side information within a session (e.g., packet length, time interval) and is typically represented by models like random forest and gradient boosting. This method is characterized by low data and model resource consumption, rapid training and prediction speeds, good real-time performance, and the use of lightweight machine learning models, making it unsuitable for extensive data processing[13,16].

Sequence-based methods within DFI technology can be categorized into three types. The first method utilizes packet length and other packet side information without statistical aggregation, employing the original feature sequence as model input, often represented by models like LSTM and one-dimensional convolutional neural networks. The second method constructs feature map data based on multiple feature correlations, represented by graph neural network models[11,13,17–19]. The third method directly converts the original byte sequence of application load data into grayscale or color images, suitable for 2D convolutional neural network models. Sequence-based methods are characterized by high data and model resource consumption, slower training and prediction speeds, and limited real-time performance. However, their use of deep learning models makes them suitable for parallel processing of large traffic data.

Hybrid inspection and identification. The limitations of DPI technology, such as the inability to identify encrypted traffic and high deployment costs, pose significant challenges in real network traffic annotation. Nonetheless, DPI technology offers advantages like simple algorithms, easy implementation, and high application identification accuracy. To address these challenges, our dataset[12] leverages a combination of DPI and DFI technologies for annotating network traffic related to service applications. While DPI technology is effective for non-encrypted traffic, manual annotation is employed for encrypted traffic, complemented by DPI annotation for non-encrypted traffic to annotate the entire dataset and develop a DFI model. Subsequently, the trained DFI model enhances the identification of unrecognized and encrypted traffic, resulting in a traffic classification identification accuracy exceeding 95%[20–22].

Related work. In recent years, the industry has acknowledged numerous public traffic datasets tailored to various field scenarios. These datasets are typically curated for specific domains and serve purposes ranging from model training for traffic classification and identification to direct statistical analysis of patterns. Prominent network traffic analysis scenarios encompass datasets like the Internet of Things (IoT) dataset, IDS series dataset, ISCX series dataset, and dark network dataset. Table 1 offers a comprehensive overview of information on seven publicly available datasets spanning four major categories, alongside the datasets introduced in this article.

ISCXVPN2016. This dataset was developed in 2016 and is mainly used to distinguish between VPN and non VPN traffic data. Developers define a set of tasks and create two simulated accounts for network communication. Based on application services such as Skype and Facebook, they simulate the generation of large categories of traffic data such as web browsing, email, chat, video streaming, file transfer, network phone, P2P transmission, etc. To ensure the purity of traffic, all unnecessary services and applications will be shut down when capturing traffic.

Filters are used to only capture packets with the source or target IP, i.e. the IP addresses of two local client accounts. The traffic was captured using Wireshark and tcpdump, resulting in a total of 28GB of data stored in Packet Capture (PCAP) file format[23]. For VPN data, use an external VPN service provider and connect using OpenVPN (UDP mode), instead of using a regular network connection for non VPN data.

ISCXTor2016. This dataset was developed in 2016 and is mainly used to distinguish between BT (Bit Torrent) and non BT traffic data. The dataset capture, feature extraction, and data annotation are similar to the

Literature	Dataset	Field	FIle Format	Size
Gerard Drapper Gil[23]	ISCXVPN2016	service application identification	PCAP	27.3 G
Arash Habibi Lashkari[24]	ISCXTor2016	service application identification	PCAP/CSV	20 G/14 M
Iman Sharafaldin[25]	CSE-CIC-IDS2018	Network attack detection	PCAP	53 G
Iman Sharafaldin[26]	CIC-DDoS 2019	Network attack detection	PCAP/CSV	21 G/3 G
Arash Habibi Lashkar[27]	CIC-Darknet2020	Abnormal traffic detection	CSV	58 M
Sajjad Dadkhah[28]	CIC IoT Profiling Dataset 2022	Behavior identification of IoT devices	PCAP/CSV	13.2 G/122 M
E. C. P. Neto[29]	CIC IoT Attack Dataset 2023	IoT device attack detection	PCAP/CSV	27.3 G/260 M
ours	—	service application identification	PCAP/CSV	10.4 G/88 M

Table 1. Overview of public datasets.

ISCXVPN2016 dataset. The ISCXTor2016 dataset contains a total of 22GB of traffic data, including web browsing, email, chat, video streaming, file transfer, network phone, P2P transmission, and other categories. In addition to being stored in the original PCAP file format, the generated PCAP file is also read by ISCxFlowMeter and corresponding CSV feature files are created based on the selected functions[24].

CSE-CIC-IDS2018. This dataset was developed in 2018 and captures traffic from Amazon e-commerce platform servers. It is generated systematically using the concept of configuration files, which will include detailed descriptions of intrusions and abstract distribution models of applications, protocols, or lower level network entities. These configuration files contain feature representations of network events and behaviors, which are combined to generate a different set of datasets, each with a unique set of features. The final dataset includes 5 different attack scenarios: brute force attacks, botnet attacks, denial of service (DDos) attacks, web attacks, and penetration attacks. The attack infrastructure includes 50 machines, and the victim organization has 5 departments, including 420 machines and 30 servers. This dataset includes network traffic captured by each machine and system logs[25]. CSE-CIC-IDS2018 is based on the CICFlowMeter tool for traffic capture and parsing. The CICFlowMeter tool is a network traffic generator written in Java. Its feature capture is based on bidirectional flow and can generate up to 83 statistical features, such as duration, number of packets, number of bytes, packet length, etc. It is calculated in two directions: upstream and downstream.

CIC-DDoS 2019. This dataset was developed in 2019 and includes both normal and the latest common DDoS attacks. By using simulated DDoS attacks, the training and testing sets were generated by executing 12 DDoS attacks and 7 DDoS attacks respectively. Each attack required recording of raw data, including network traffic (Pcaps) and event logs for each machine. Then, the CICFlowMeter tool was used for network traffic analysis, extracting over 80 traffic features and saving them as CSV files for each machine, If you want to extract custom features, you can use the original PCAP file to extract feature[26]. This dataset records the behavior data of 25 users based on application layer email protocols such as HTTP, HTTPS, FTP, SSH, etc. The vulnerable protocols used for attacks include NTP, DNS, LDAP, MSSQL, BIOS, SNMP, SSDP, UDP, UDP Lag, WebDDoS, SYN, and TFTP.

CIC-Darknet2020. This data set was developed in 2020. The dark network comes from the unused IP address space of the Internet. Any traffic data from the dark network has a passive listening nature, accepting incoming packets, but not supporting outgoing packets. Due to the absence of legitimate hosts in dark networks, any traffic cannot be traced, and is often considered a network testing or configuration error. The classification of dark network traffic is very important for classifying real-time applications, and analyzing dark network traffic helps to monitor malware and detect malicious activities early before malware attacks[27]. CIC-Darknet2020 merged two other public datasets, ISCxTor2016 and ISCxVPN2016, to detect and characterize VPN and Tor applications together as the true representatives of dark network traffic, in order to create a complete dark network dataset that covers Tor and VPN traffic, respectively. In the CICDarknet2020 dataset, traffic categories include audio streaming, browsing, chatting, email, P2P, transmission, video streaming, and VOIP, with each category containing both normal and dark network traffic.

CIC IoT profiling dataset 2022. This dataset was developed in 2022 with the main goal of generating a comprehensive IoT dataset for analysis, behavior analysis, and vulnerability testing of different IoT devices using different protocols (such as IEEE 802.11, based on Zigbee and Z-Wave). This dataset configures various network traffic data for conducting and capturing device undercurrents and important attacks in IoT environments, analyzes the behavior exhibited by IoT devices, and includes different scenarios of IoT and analyzes the behavior of devices in different situations. In addition, this dataset also includes traffic data of IoT devices during startup and idle periods, making it convenient for analyzing specific scenarios. The traffic data collection environment is configured with a 64 bit Window computer through laboratory network configuration, which has two network interface cards and one connected to a network gateway and the other connected to an unmanaged network switch. Meanwhile, the open-source network protocol analyzer Wireshark listens to these two interfaces, capturing and saving output pcap files[28].

Therefore, IoT devices that require Ethernet connection will be connected to this switch. In addition, the intelligent automation center Vera Plus is also connected to a non managed switch, which creates our wireless IoT environment and provides services for IoT devices compatible with Wi Fi, ZigBee, Z-Wave, and Bluetooth. In order to collect data, Wireshark and dumpcap were used in six different types of experiments to capture the

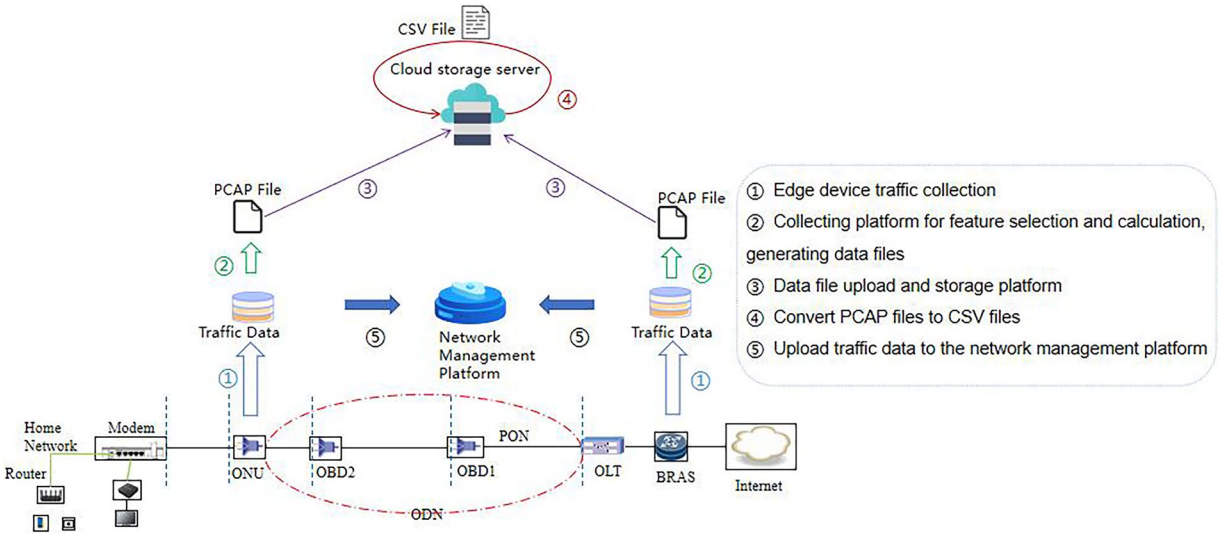


Fig. 1 Data collection framework.

network traffic of IoT devices passing through the gateway, generating PCAP files and dumpcap files, which can be conveniently used for various model experiments related to IoT application scenarios. This dataset category covers four different scenarios: startup/shutdown comparison traffic data of IoT devices, active/idle comparison traffic data of devices in startup state, traffic data of devices communicating with each other, and traffic data of devices being attacked.

CIC IoT attack dataset 2023. This dataset was developed in 2023 and can serve as a real-time benchmark dataset for large-scale attacks in IoT environments. Its main goal is to provide a novel and extensive IoT attack dataset to promote the development of security analysis applications in actual IoT operations[29]. To this end, 105 attacks were executed in an IoT topology consisting of 33 devices. The environment, method, and generated data files for traffic data collection are similar to the CIC IoT Profiling Dataset 2022 dataset.

Our dataset. Existing public datasets are often tailored to specific applications in particular scenarios, such as IoT data and network attack data, resulting in fragmented coverage and limited scope. These datasets are typically generated in laboratory settings for simulated data collection, influenced by the hardware environment and experimental configurations. The controlled settings lead to small-scale data collection, restricting the representation of real-world service applications accessed by users. Moreover, public datasets offer static data from a limited number of test users, lacking updates and timeliness.

In contrast, the data collection approach proposed in this article involves gathering data from large-scale edge devices in authentic network environments, primarily focusing on access devices. By deploying plugins or platforms on these access devices, real user internet traffic is collected, providing a more accurate reflection of the diverse service applications accessed by users. The dataset[12] introduced in this article can effectively identify and classify both encrypted and non-encrypted traffic, enabling operators to tailor precise marketing strategies. Additionally, it facilitates the provision of customized proactive services for different user segments, ultimately enhancing user satisfaction.

Methods

This paper focuses on creating real Internet data, and the focus of this dataset[12] is data collection and data feature extraction. This article will first introduce the overall method of generating traffic datasets based on the data collection sources of different network location segments, and then subdivide the content into three parts: data collection, data processing, and online deployment.

Overall method for generating datasets. The data collection framework encompasses two primary components: traffic acquisition from access devices, specifically the Broadband Remote Access Server (BRAS), and edge devices, particularly the Optical Network Unit (ONU), as shown in Fig. 1. Within the network topology, the BRAS and ONU serve as critical elements for traffic monitoring and data acquisition.

The BRAS, functioning as the access point for the Optical Line Terminal (OLT) port, plays a pivotal role in optical signal reception and transmission. Typically interconnected with multiple network devices through an optical brancher, it serves as the crucial interface bridging user equipment and the operator's core network infrastructure. Through its integration with the OLT, the BRAS executes essential operations including optical multiplexing/demultiplexing, channel connection assignment and control, while simultaneously providing auxiliary functionalities such as real-time network monitoring, system management, and maintenance operations.

The network architecture primarily consists of two essential passive components: the Optical Distribution Network (ODN) and its integrated Optical Branching Device (OBD). The OBD features a hierarchical structure comprising two distinct branches: the primary Optical Branching Device (OBD1) and the secondary Optical

LOID	OBD2 code	OBD1 code	OLT code	OLT port	BRAS code	BRAS port
13257376	GW025015/OBD7005/OUT02	GW025/OBD8011	NC-NB-DongF2F-OLT.HXTA6800X15-01(09)	EPON_1/6/2	SC-NC-NCDGL-NSE-1.MAN.TE40E	D02/K/TE40E-X16-0001/01/06.0(P101E)/01/GigabitEthernet3/1/1

Table 2. An example of Network topology resource tree information.

Branching Device (OBD2), which collectively facilitate optical signal distribution within the network infrastructure. At the user terminal, the home gateway serves as the output interface of the PON port, primarily functioning as the user interface between terminal equipment and the Passive Optical Network (PON). Beyond its fundamental role in optoelectronic conversion (supporting Ethernet frame format), the gateway implements comprehensive network functionalities, including Layer 2 and Layer 3 switching operations. These advanced capabilities facilitate efficient data routing between network segments, thereby enhancing transmission reliability and network performance.

When numerous network units like access devices and edge devices continuously monitor Internet traffic, they can capture real-time packets from a large volume of users utilizing the network simultaneously. To effectively leverage this data, a series of processing and transformations are necessary. Initially, the plugin system deployed in the network units conducts data cleaning to eliminate erroneous or abnormal data, ensuring the accuracy of subsequent processing. The plugin system processes the captured traffic data by filtering and converting formats to enhance its suitability for further analysis.

Following data cleaning and feature calculation, the data undergoes additional processing to generate PCAP and CSV files. PCAP files, a standard format for capturing network traffic, meticulously record packet information flowing through network units, serving as raw material for subsequent traffic analysis. Conversely, CSV files present data in tabular form, facilitating rapid data analysis and processing by researchers. These files are stored in a cloud server to ensure data accessibility and persistence[30,31].

Massive data files are downloaded from the cloud server to the local server through an interface between the two. On the local server, this data is utilized to train AI models, enabling them to learn network traffic characteristics and patterns, analyze data using proprietary algorithms and model structures. Through continuous learning and refinement, the model's prediction accuracy improves gradually, uncovering valuable insights and patterns. During the training process, the AI model offers feedback for optimizing plugin deployment based on prediction results and feature selection methods. Upon receiving this feedback, the plugin system guides plugins to enhance and adjust feature calculation methods. This closed-loop data processing mechanism ensures continuous system optimization and progress. Through this collaborative approach, network units, plugin systems, cloud servers, and local servers collectively form an efficient and intelligent data processing system, enhancing data utilization and providing robust support for network traffic analysis.

To validate the authenticity of the data acquisition environment, Table 2 presents an actual network topology resource tree link information, including the Logical Identification (LOID) of an user's optical modem, OBD2 code, OBD1 code, OLT code, OLT port, BRAS code, BRAS port.

Data collection. *Data collection from edge devices.* The ONU plug-in collects raw traffic based on port replication. Since some session flows contain a large number of data payload packets, and considering the usage of computing and storage resources deployed online, only a portion of the packets are selected for statistical computing[32]. Therefore, the first N packets of each flow are intercepted for statistical feature extraction. experimental methods were used to select $N = 4,6,8,12,16,24,32,48,64$. According to the trend chart of the accuracy rate of classification and recognition based on business application categories, when $N < 8$, increasing the value of N causes the classification accuracy rate to increase rapidly, However, after $N > 8$, the model classification accuracy tends to remain almost unchanged. It is finally determined that it is most appropriate to collect only the first 8 (the number of packets can be set remotely by the perception platform) packets for statistical feature extraction through three layers of switching. The relationship between the number of packets per stream and the service classification accuracy is shown in Fig. 2.

By utilizing the traffic capture engine within the perception plugin to capture packets from specified network interfaces, protocol analysis is conducted on network packets. Data sharing the same IP + port + layer 4 protocol (5-tuple) is categorized as a stream, enabling in-depth analysis of the data stream to generate corresponding logs for streams conforming to protocol rules. Integration of perception plugins into gateway devices enables comprehensive traffic data collection, with the plugin monitoring designated links as per platform directives to assess network quality and interface with the network quality monitoring platform[33].

The process commences with the network management platform issuing specific traffic data collection instructions, generating subscription tasks through the southbound interface to interact with the service data center platform and the USM plugin data platform. The subscription command is dispatched to the designated terminal, where the plugin receives and processes the subscription request. The raw traffic byte sequence is read, and the service data center platform collaborates with the backend data platform to parse, extract features, and annotate the application protocol of the raw data byte streams in batches.

Post-parsing of the original traffic, data such as five tuples, MAC, HOST, User Agent, Cookies, SNI, packet counts, packet intervals, and packet sizes are obtained. The backend analysis platform categorizes and organizes diverse data based on services, discarding non-client-server communication packets lacking internal network addresses. This organized data is periodically relayed to the network management platform in a predefined format, with a default reporting interval of 100 ms, remotely adjustable via the awareness platform.

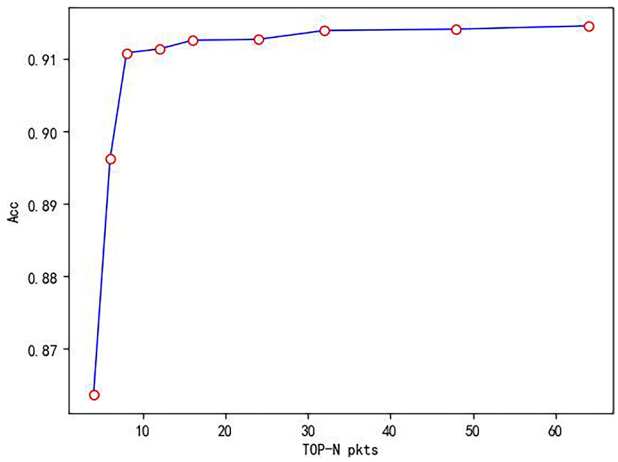


Fig. 2 The relationship between top-n packets of each flow and bussiness recognition.

The edge devices plugin processes the reported data by extracting, cleaning, filtering, labeling, and removing duplicates, enhancing subsequent data analysis and processing capabilities. Leveraging host, URL, and other reported information, a high-performance application identification engine based on Hyperscan is utilized to identify user applications and construct a service identification feature library[34]. The application feature library is then disseminated to the plugin, enabling matching of message feature words or features to ascertain traffic service types. Ultimately, the five tuples, side channel features (packet quantity, length, arrival time interval), and corresponding service application labels are utilized for application classification and identification. Additionally, IP and application libraries are established to support data classification and identification.

Data collection from access devices. To implement traffic data collection on a access devices device in a prefecture-level city, with 19,417 online users generating bidirectional traffic of around 45 Gbit/s during peak hours from 8 to 10 in the evening (Beijing Standard Time Zone, China), and approximately 2.47 million network session connections every 5 minutes, the following steps can be taken[35]:

- 1. Deploy software packages on the access devices device to facilitate traffic data collection.
- 2. Configure the software to collect traffic data from multiple users of broadband web applications.
- 3. Set parameters such as the maximum packet capture rate, number of packets, duration, and capture mode (raw or pure stream capture) for manual packet capture.
- 4. Upon completion of the capture process, the interface should automatically generate PCAP files for further analysis. For traffic application protocol matching identification on the access devices device, deep packet detection technology is employed with two distinct methods:

The BRAS traffic application protocol matching and identification method employs deep packet inspection technology, which can be implemented in two approaches:

- 1. DNS Domain Name Resolution Method: Extract application and protocol names from DNS domain name resolutions of various HTTP protocol-based fields (e.g., host, user-agent) by tracking HTTP flows.
- 2. Payload Byte Sequence Method: Search for specific byte sequences within a large volume of original payload traffic to establish a mapping rule table between the payload byte sequences and application protocols. - Maintain this mapping rule table to identify newly captured traffic application protocols based on the protocol rule table.

Data processing. The traffic data collected by edge devices and access devices undergoes processing by the plugin platform or board system to generate data files (PCAP, CSV) that are then uploaded to the cloud storage server. The workflow involves uploading PCAP files to the ownCloud server, which are subsequently transferred to the local server. Conversely, CSV files are initially uploaded to the local server, where all PCAP files are converted into CSV format. The CSV files from various sources are consolidated and fed into the AI model for feature selection. The offline AI model provides feedback on feature importance scores to the edge devices and access devices collection platforms for re-selection and calculation of traffic features. This process, depicted in Fig. 1, encompasses steps 3–5 as a closed loop for traffic feature selection. Upon finalizing the essential features, the results are stored in the ownCloud cloud storage server in a stable CSV format as the definitive traffic data. The extracted traffic features typically fall into three categories[36]:

- 1. Dimensional Features: Comprising five key features: user-server IP, user-server port number, and protocol. These five-tuple features uniquely identify session basic information.

Application category	Application instance
network-storage	Baidu Netdisk
	Tianyi Cloud Disk
	Alibaba Cloud
	Hua Weiyun
network-transmission	Thunderbolt
	BT Download
	Emule
video	Station B
	Tiktok
	Tencent Video
	IQiyi Video
	Youku Video
	Mango TV
	Tencent Meeting
game	League of Legends
	Honor of Kings
instant-message	QQ
	WeChat
web-browsing	Alipay webpage
	Today's Headlines
	Zhihu
	Baidu Baike
mail-service	163 email
	QQ email
	189 email

Table 3. Service application classification label.

- 2. Statistical Features: Utilized for identifying side channel information in DFI technology, including session packet size, packet arrival time interval, protocol features, etc.
- 3. Label Features: Encompassing application categories and specific sub-applications corresponding to the conversation, aiding in traffic classification and identification.

Service application classification label. Based on the statistical distribution analysis of real network users' access to Internet application instances and leveraging traffic monitoring professional software, a set of nine standardized service application category labels has been established. These categories include network storage, network transmission, network video, web games, instant messaging, network security, tunnel protocol, mail service, and web browsing[1,3]. The selection criteria for these applications prioritize mainstream popular applications that are widely accessed by real network users. This selection approach ensures that the collected data is not only easily obtainable but also more indicative of the traffic characteristics associated with each category tag. For a detailed breakdown of application category names and specific examples from the real network traffic data selection, please refer to Table 3.

Service application feature extraction. The feature extraction process for real network data involves several steps, as outlined in Fig. 3:

- 1. Pure Purification Treatment of Original PCAP Files: - Split the PCAP file into different sessions based on the same quintuple and use the SplitCap tool to divide the original captured PCAP file into multiple pure stream files, each containing a unique quintuple.
- 2. Invalid IP Address Removal: - Eliminate session flows lacking user IPs, typically intranet or special addresses like 192.168.x.x or 100.64.x.x.
- 3. Packet Filtering: - Select only the first 8 data packets for each session to optimize resource deployment for feature extraction algorithms.
- 4. Packet Division: - Divide each packet into uplink and downlink packets based on whether the user IP is the source or destination, resulting in three types of packet sets.
- 5. Original Feature Selection: - Extract two sets of side channel numerical sequences from each data packet: packet arrival time interval and packet byte number.
- 6. Feature Statistical Calculation: - Compute 8 statistical values for each session flow packet, including minimum, maximum, mean, median absolute deviation, standard deviation, median, 25th percentile, and 75th percentile. This results in a total of 48 features across uplink packets, downlink packets, and all packets.

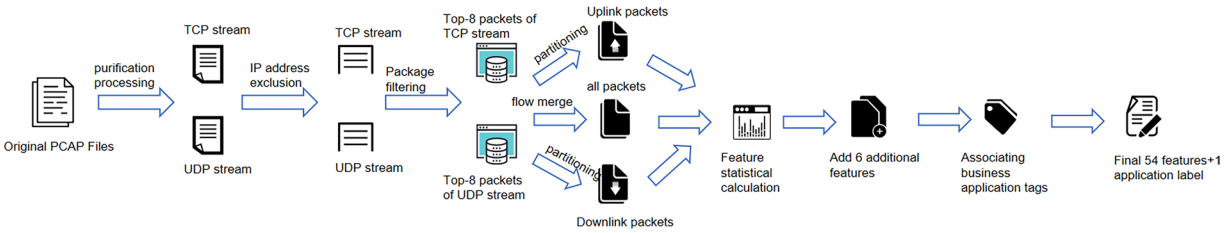


Fig. 3 Feature extraction process of real network data.

Flow direction	Side channel information	Feature Count	Feature representation (s = min,max,mean,mad,std,p25,p50,p75)
double-link	Package arrival time interval	8	$T_{\Delta s}$
	Package length	8	$L_{\Delta s}$
	Transport layer protocol encoding	1	C_{proto}
uplink	Package arrival time interval	8	$T^{up}_{\Delta,s}$
	Package length	8	$L^{up}_{\Delta,s}$
	Number of packages	3	N^{up}_{http}, N^{up}_{https}, N^{up}
downlink	Package arrival time interval	8	$T^{down}_{\Delta,s}$
	Package length	8	$L^{down}_{\Delta,s}$
	Number of packages	2	N^{down}_{http}, N^{down}_{https}

Table 4. Extracted statistical features.

- 7. Increase the number of Packets Feature: - Enhance the feature set by including the number of uplink packets.
- 8. Add Transport Layer Protocol Features: - Introduce transport layer protocol encoding (0 for TCP, 1 for UDP protocol, otherwise 2).
- 9. Add Application Layer Protocol Features: - Include the number of fixed ports 80/443 corresponding to HTTP/HTTPS, resulting in 4 features for uplink and downlink.
- 10. Associated Service Application Labels:For edge devices data, extract and associate with DPI, match with the IP and port of the PCAP file based on the server IP and port to label each row of feature data. For access devices data, the service application name in the PCAP file directly associates with each row of feature data.

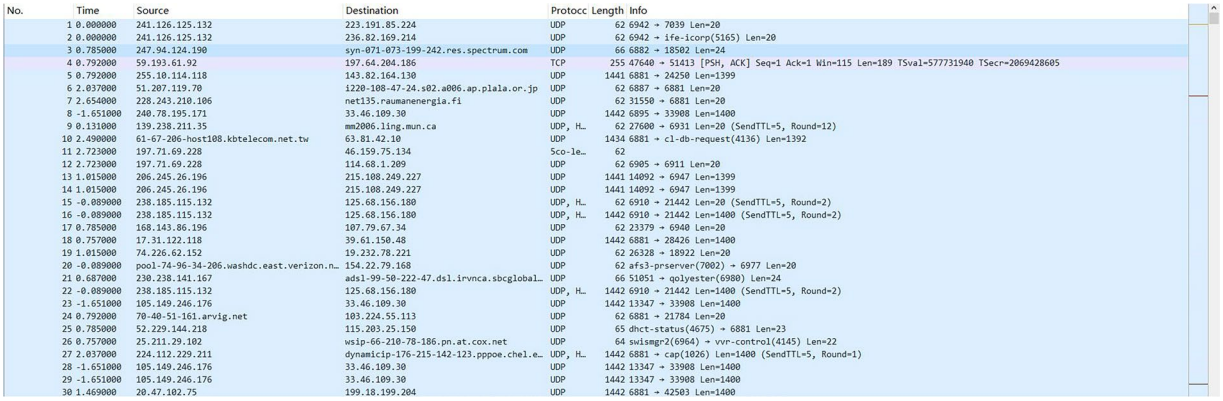
The 54 feature representation segments are detailed in Table 4. The features are represented as: $T^{d}_{\Delta,s}$, $L^{d}_{\Delta,s}$, N^{d}_{proto}, C_{proto}, where *T, L, N*, and *C* denote packet inter-arrival time, packet length, packet count, and protocol code, respectively. The superscript *d* indicates the traffic direction: $d = up,down$. The absence of the superscript *d* indicates bidirectional traffic. For the *T* and *L* features, the subscript Δ denotes the variation in time or length, and the subscript *s* represents the statistical feature state: $s = min,max,mean,mad,std,p25,p50,p75$, which stand for minimum, maximum, mean, median absolute deviation, standard deviation, 25th percentile, 50th percentile (median), and 75th percentile, respectively. For the N feature, the subscript *proto* indicates the selection of the application layer protocol. If this subscript is absent, it means that no specific protocol is distinguished Figs. 4, 5.

Data Records

The proposed dataset[12] is publicly available in the figshare repository (https://doi.org/10.6084/m9.figshare.28380347). It comprises four files, including two archived files containing network traffic data and two CSV files with metadata. Notably, all PCAP files within the original compressed traffic archives have undergone anonymization of both source and destination MAC/IP addresses to ensure privacy compliance:

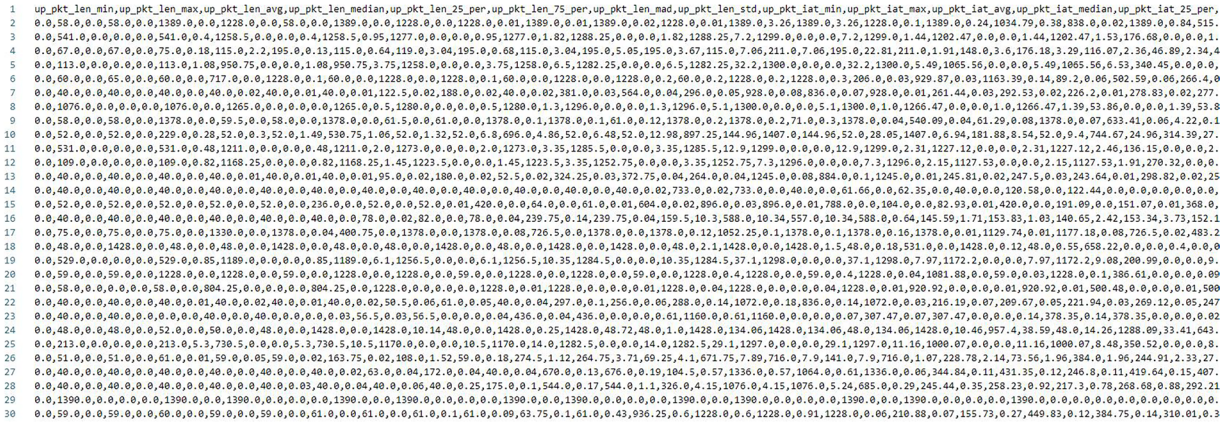
- 1. BRAS-datasets.rar: Contains raw traffic data (PCAP format) captured from BRAS network elements, encompassing 23 data files across 7 service application categories described in this study (Total size: 5.36 GB).
- 2. ONU-datasets.rar: Includes raw traffic data (PCAP format) collected from ONU network units, containing 41 data files spanning 7 service application categories specified in this work (Total size: 5.05 GB).
- 3. Bras_features.csv: Feature file generated through our proposed methodology, containing extracted characteristics from BRAS-originated PCAP files.
- 4. Onu_features.csv: Feature file derived using our presented approach, comprising processed features from ONU-sourced PCAP files.

Additionally, both raw datasets collected from edge devices (ONU) and access devices (BRAS), along with feature-engineered datasets for model input, are publicly accessible via GitHub: https://github.com/zqxfree/Real-network-datasets-for-application-analysis/blob/main/dataset. Our dataset[12] records encompass the following 8 aspects:



No.	Time	Source	Destination	Protocc	Length	Info
1	0.000000	241.126.125.132	223.191.85.224	UDP	62	6942 → 7039 Len=20
2	0.000000	241.126.125.132	236.82.169.214	UDP	62	6942 → ife-icorp(5165) Len=20
3	0.785000	247.94.124.190	syn-071-073-199-242.res.spectrum.com	UDP	66	6882 → 18502 Len=24
4	0.792000	59.193.61.92	197.64.204.186	TCP	255	47640 → 51413 [PSH, ACK] Seq=1 Ack=1 Win=115 Len=189 TSval=577731940 TSecr=2069428605
5	0.792000	255.10.114.118	143.82.164.130	UDP	1441	6881 → 24250 Len=1399
6	2.037000	51.207.119.70	1220-108-47-24.s02.a006.ap.plala.or.jp	UDP	62	6887 → 6881 Len=20
7	2.654000	228.243.210.106	net135.raumanenergia.fi	UDP	62	31550 → 6881 Len=20
8	-1.651000	240.78.195.171	33.46.109.30	UDP	1442	6895 → 33908 Len=1400
9	0.131000	139.238.211.35	mm2006.ling.mun.ca	UDP, H…	62	27600 → 6931 Len=20 (SendTTL=5, Round=12)
10	2.490000	61-67-206-host108.kbtelecom.net.tw	63.81.42.10	UDP	1434	6881 → cl-db-request(4136) Len=1392
11	2.723000	197.71.69.228	46.159.75.134	Sco-le…	62	
12	2.723000	197.71.69.228	114.68.1.209	UDP	62	6905 → 6911 Len=20
13	1.015000	206.245.26.196	215.108.249.227	UDP	1441	14092 → 6947 Len=1399
14	1.015000	206.245.26.196	215.108.249.227	UDP	1441	14092 → 6947 Len=1399
15	-0.089000	238.185.115.132	125.68.156.180	UDP, H…	62	6910 → 21442 Len=20 (SendTTL=5, Round=2)
16	-0.089000	238.185.115.132	125.68.156.180	UDP, H…	1442	6910 → 21442 Len=1400 (SendTTL=5, Round=2)
17	0.785000	168.143.86.196	107.79.67.34	UDP	62	23379 → 6940 Len=20
18	0.757000	17.31.122.118	39.61.150.48	UDP	1442	6881 → 28426 Len=1400
19	1.015000	74.226.62.152	19.232.78.221	UDP	62	26328 → 18922 Len=20
20	-0.089000	pool-74-96-34-206.washdc.east.verizon.n…	154.22.79.168	UDP	62	afs3-prserver(7002) → 6977 Len=20
21	0.687000	230.238.141.167	ads1-99-50-222-47.dsl.irvnca.sbcglobal…	UDP	66	51051 → qolyester(6980) Len=24
22	-0.089000	238.185.115.132	125.68.156.180	UDP, H…	1442	6910 → 21442 Len=1400 (SendTTL=5, Round=2)
23	-1.651000	105.149.246.176	33.46.109.30	UDP	1442	13347 → 33908 Len=1400
24	0.792000	70-40-51-161.arvig.net	103.224.55.113	UDP	62	6881 → 21784 Len=20
25	0.785000	52.229.144.218	115.203.25.150	UDP	65	dhct-status(4675) → 6881 Len=23
26	0.757000	25.211.29.102	wsip-66-210-78-186.pn.at.cox.net	UDP	64	swismgr2(6964) → vvr-control(4145) Len=22
27	2.037000	224.112.229.211	dynamicip-176-215-142-123.pppoe.chel.e…	UDP, H…	1442	6881 → cap(1026) Len=1400 (SendTTL=5, Round=1)
28	-1.651000	105.149.246.176	33.46.109.30	UDP	1442	13347 → 33908 Len=1400
29	-1.651000	105.149.246.176	33.46.109.30	UDP	1442	13347 → 33908 Len=1400
30	1.469000	20.47.102.75	199.18.199.204	UDP	1442	6881 → 42503 Len=1400

Fig. 4 The samples of raw data (PCAP file).



```
up_pkt_len_min,up_pkt_len_max,up_pkt_len_avg,up_pkt_len_median,up_pkt_len_25_per,up_pkt_len_75_per,up_pkt_len_mad,up_pkt_len_std,up_pkt_iat_min,up_pkt_iat_max,up_pkt_iat_avg,up_pkt_iat_median,up_pkt_iat_25_per,
0.0,58.0,0.0,58.0,0.0,1389.0,0.0,1228.0,0.0,58.0,0.0,1389.0,0.0,1228.0,0.0,1228.0,0.01,1389.0,0.01,1389.0,0.02,1228.0,0.01,1389.0,3.26,1389.0,3.26,1228.0,0.1,1389.0,0.24,1034.79,0.38,838.0,0.02,1389.0,0.84,515.
0.0,541.0,0.0,0.0,0.0,541.0,0.4,1258.5,0.0,0.0,0.4,1258.5,0.95,1277.0,0.0,0.0,0.95,1277.0,1.82,1288.25,0.0,0.0,1.82,1288.25,7.2,1299.0,0.0,0.0,7.2,1299.0,1.44,1202.47,0.0,0.0,1.44,1202.47,1.53,176.68,0.0,0.0,1.
0.0,67.0,0.0,67.0,0.0,75.0,0.18,115.0,2.2,195.0,0.13,115.0,0.64,119.0,3.04,195.0,0.68,115.0,3.04,195.0,5.05,195.0,3.67,115.0,7.06,211.0,7.06,195.0,22.81,211.0,1.91,148.0,3.6,176.18,3.29,116.07,2.36,46.89,2.34,4
0.0,113.0,0.0,0.0,0.0,113.0,1.08,950.75,0.0,0.0,1.08,950.75,3.75,1258.0,0.0,0.0,3.75,1258.0,6.5,1282.25,0.0,0.0,6.5,1282.25,32.2,1300.0,0.0,0.0,32.2,1300.0,5.49,1065.56,0.0,0.0,5.49,1065.56,6.53,340.45,0.0,0.0,
0.0,60.0,0.0,65.0,0.0,60.0,0.0,717.0,0.0,1228.0,0.1,60.0,0.0,1228.0,0.0,1228.0,0.1,60.0,0.0,1228.0,0.0,1228.0,0.2,60.0,0.2,1228.0,0.2,1228.0,0.3,206.0,0.03,929.87,0.03,1163.39,0.14,89.2,0.06,502.59,0.06,266.4,0
0.0,40.0,0.0,40.0,0.0,40.0,0.02,40.0,0.01,40.0,0.01,122.5,0.02,188.0,0.02,40.0,0.02,381.0,0.03,564.0,0.04,296.0,0.05,928.0,0.08,836.0,0.07,928.0,0.01,261.44,0.03,292.53,0.02,226.2,0.01,278.83,0.02,277.
0.0,1076.0,0.0,0.0,0.0,1076.0,0.0,1265.0,0.0,0.0,0.0,1265.0,0.5,1280.0,0.0,0.0,0.5,1280.0,1.3,1296.0,0.0,0.0,1.3,1296.0,5.1,1300.0,0.0,0.0,5.1,1300.0,1.0,1266.47,0.0,0.0,1.0,1266.47,1.39,53.86,0.0,0.0,1.39,53.8
0.0,58.0,0.0,58.0,0.0,1378.0,0.0,59.5,0.0,58.0,0.0,1378.0,0.0,61.5,0.0,61.0,0.0,1378.0,0.1,1378.0,0.1,61.0,0.12,1378.0,0.2,1378.0,0.2,71.0,0.3,1378.0,0.04,540.09,0.04,61.29,0.08,1378.0,0.07,633.41,0.06,4.22,0.1
0.0,52.0,0.0,52.0,0.0,229.0,0.28,52.0,0.3,52.0,1.49,530.75,1.06,52.0,1.32,52.0,6.8,696.0,4.86,52.0,6.48,52.0,12.98,897.25,144.96,1407.0,144.96,52.0,28.05,1407.0,6.94,181.88,8.54,52.0,9.4,744.67,24.96,314.39,27.
0.0,531.0,0.0,0.0,0.0,531.0,0.48,1211.0,0.0,0.0,0.48,1211.0,2.0,1273.0,0.0,0.0,2.0,1273.0,3.35,1285.5,0.0,0.0,3.35,1285.5,12.9,1299.0,0.0,0.0,12.9,1299.0,2.31,1227.12,0.0,0.0,2.31,1227.12,2.46,136.15,0.0,0.0,2.
0.0,109.0,0.0,0.0,0.0,109.0,0.82,1168.25,0.0,0.0,0.82,1168.25,1.45,1223.5,0.0,0.0,1.45,1223.5,3.35,1252.75,0.0,0.0,3.35,1252.75,7.3,1296.0,0.0,0.0,7.3,1296.0,2.15,1127.53,0.0,0.0,2.15,1127.53,1.91,270.32,0.0,0.
0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.01,40.0,0.01,40.0,0.01,95.0,0.02,180.0,0.02,52.5,0.02,324.25,0.03,372.75,0.04,264.0,0.04,1245.0,0.08,884.0,0.1,1245.0,0.01,245.81,0.02,247.5,0.03,243.64,0.01,298.82,0.02,25
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0.0,52.0,0.0,52.0,0.0,52.0,0.0,52.0,0.0,52.0,0.0,236.0,0.0,52.0,0.0,52.0,0.0,52.0,0.01,420.0,0.0,64.0,0.0,61.0,0.01,604.0,0.02,896.0,0.03,896.0,0.01,788.0,0.0,104.0,0.0,82.93,0.01,420.0,0.0,191.09,0.0,151.07,0.01,368.0,
0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.0,78.0,0.02,82.0,0.0,78.0,0.04,239.75,0.14,239.75,0.04,159.5,10.3,588.0,10.34,557.0,10.34,588.0,0.64,145.59,1.71,153.83,1.03,140.65,2.42,153.34,3.73,152.1
0.0,75.0,0.0,75.0,0.0,75.0,0.0,1330.0,0.0,1378.0,0.04,400.75,0.0,1378.0,0.0,1378.0,0.08,726.5,0.0,1378.0,0.0,1378.0,0.12,1052.25,0.1,1378.0,0.1,1378.0,0.16,1378.0,0.01,1129.74,0.01,1177.18,0.08,726.5,0.02,483.2
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0.0,40.0,0.0,40.0,0.0,40.0,0.01,40.0,0.02,40.0,0.01,40.0,0.02,50.5,0.06,61.0,0.05,40.0,0.04,297.0,0.1,256.0,0.06,288.0,0.14,1072.0,0.18,836.0,0.14,1072.0,0.03,216.19,0.07,209.67,0.05,221.94,0.03,269.12,0.05,247
0.0,40.0,0.0,40.0,0.0,0.0,0.0,40.0,0.0,40.0,0.0,0.0,0.0,3.56,5.0,3.56,5.0,0.0,0.0,0.04,436.0,0.04,436.0,0.0,0.0,0.0,0.61,1160.0,0.61,1160.0,0.0,0.0,0.0,0.07,307.47,0.07,307.47,0.0,0.0,0.14,378.35,0.14,378.35,0.0,0.0,0.02
0.0,48.0,0.0,48.0,0.0,52.0,0.0,50.0,0.0,48.0,0.0,1428.0,0.0,1428.0,10.14,48.0,0.0,1428.0,0.25,1428.0,48.72,48.0,1.0,1428.0,134.06,1428.0,134.06,48.0,134.06,1428.0,10.46,957.4,38.59,48.0,14.26,1288.09,33.41,643.
0.0,213.0,0.0,0.0,0.0,213.0,5.3,730.5,0.0,0.0,5.3,730.5,10.5,1170.0,0.0,0.0,10.5,1170.0,14.0,1282.5,0.0,0.0,14.0,1282.5,29.1,1297.0,0.0,0.0,29.1,1297.0,11.16,1000.07,0.0,0.0,11.16,1000.07,8.48,350.52,0.0,0.0,8.
0.0,51.0,0.0,51.0,0.0,61.0,0.01,59.0,0.05,59.0,0.02,163.75,0.02,108.0,1.52,59.0,0.18,274.5,1.12,264.75,3.71,69.25,4.1,671.75,7.89,716.0,7.9,141.0,7.9,716.0,1.07,228.78,2.14,73.56,1.96,384.0,1.96,244.91,2.33,27.
0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.02,63.0,0.04,172.0,0.04,40.0,0.04,670.0,0.13,676.0,0.19,104.5,0.57,1336.0,0.57,1064.0,0.61,1336.0,0.06,344.84,0.11,431.35,0.12,246.8,0.11,419.64,0.15,407.
0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.0,40.0,0.03,40.0,0.04,40.0,0.06,40.0,0.25,175.0,0.1,544.0,0.17,544.0,1.1,326.0,4.15,1076.0,4.15,1076.0,5.24,685.0,0.29,245.44,0.35,258.23,0.92,217.3,0.78,268.68,0.88,292.21
0.0,1390.0,0.0,0.0,0.0,1390.0,0.0,1390.0,0.0,0.0,0.0,1390.0,0.0,1390.0,0.0,0.0,0.0,1390.0,0.0,1390.0,0.0,0.0,0.0,1390.0,0.0,1390.0,0.0,0.0,0.0,1390.0,0.0,1390.0,0.0,0.0,0.0,1390.0,0.0,1390.0,0.0,0.0,0.0,1390.0,0.0,0.0,0.0,0.0,0.0,0.0,0.
0.0,59.0,0.0,59.0,0.0,60.0,0.0,59.0,0.0,59.0,0.0,61.0,0.0,61.0,0.0,61.0,0.0,1,61.0,0.09,63.75,0.1,61.0,0.43,936.25,0.6,1228.0,0.6,1228.0,0.91,1228.0,0.06,210.88,0.07,155.73,0.27,449.83,0.12,384.75,0.14,310.01,0.3
```

Fig. 5 The samples of processed data after feature extraction (CSV file).

- 1. Traffic Data Source: The traffic data is sourced from two collection platforms: access devices collection boards and edge devices collection plugins, capturing real user traffic from the network.
- 2. Data Application Scope: The data is utilized for traffic analysis and service application identification within the network environment.
- 3. Data File Format: Data collected by the access devices and edge devices plugins are all stored in PCAP file format.
- 4. Collection Environment: The access devices collection platform is situated on a device serving 19.5 thousand active users with 45Gbit/s bidirectional traffic during peak hours. The edge devices plugin is deployed on an edge devices in a prefecture-level city, generating 400 log entries per unit per second, serving 26.4 thousand active users with a daily storage capacity of 3.3TB in the ES database.
- 5. Data Volume: The access devices platform gathers 5.36GB of service application identification data also in PCAP format while the edge devices plugin platform collects 5.05GB of data stored in PCAP files.
- 6. Data Collection Period (Beijing Standard Time Zone, China): Data collection spans morning (9:00–12:00), afternoon (14:00–17:00), and evening (18:00–22:00) time periods.
- 7. Selected Features: A total of 54 features are extracted from the access devices plugins and edge devices boards for model input, with standardized service application category labels as model output.
- 8. Feature Extraction: The access devices plugin and edge devices board data generate 244,216 and 81,406 samples, respectively, after feature extraction.

These aspects provide a comprehensive overview of the data collection process, sources, formats, environment, volume, features, and collection period for the traffic data obtained from edge devices and access devices devices[19–21] Table 5.

Technical Validation

This section initially compares our extracted dataset[12] with a publicly available one, ISCXVPN2016, to verify its data integrity, quality, and authenticity, as depicted in Fig. 6. We choose ISCXVPN2016, which closely resembles the dataset[12] proposed in this paper, for the comparison. Here are the three aspects of comparison:

source	Total size	bandwidth	Collection period	Original data format	Processed file format	Total session number	Total packet number	Precessing Method	Model input features	Model output features (labels)	CSV file line count	Sample correspondence
edge devices (ONU)	5.05GB	22MB	9:00-12:00 14:00-17:00 18:00-22:00	PCAP	CSV	81406	10234865	Renaming PCAP files using application tags based on timestamps	Table 4	Table 3	81406	One row of data corresponds to one session and one sample
access devices (BRAS)	5.36GB	66MB	9:00-12:00 14:00-17:00 18:00-22:00	PCAP	CSV	244216	27595448	Split the original PCAP file by session	Table 4	Table 3	244216	One row of data corresponds to one session and one sample

Table 5. Overview of dataset[12] indicators.

Application category	size(GB)		session number	
	Our datasets[12]	ISCXVPN2016[23]	Our datasets[12]	ISCXVPN2016[23]
network-storage	1.20	3.12	37236	64644
network-transmission	1.87	4.85	57827	100388
video	0.38	0.97	11525	20001
game	0.47	1.21	14434	25058
instant-message	1.85	4.8	57234	99359
web-browsing	2.40	6.22	74142	128706
mail-service	2.24	6.14	73224	127118
	10.41	27.31	325622	565274

Table 6. Data quality statistical distribution comparison.

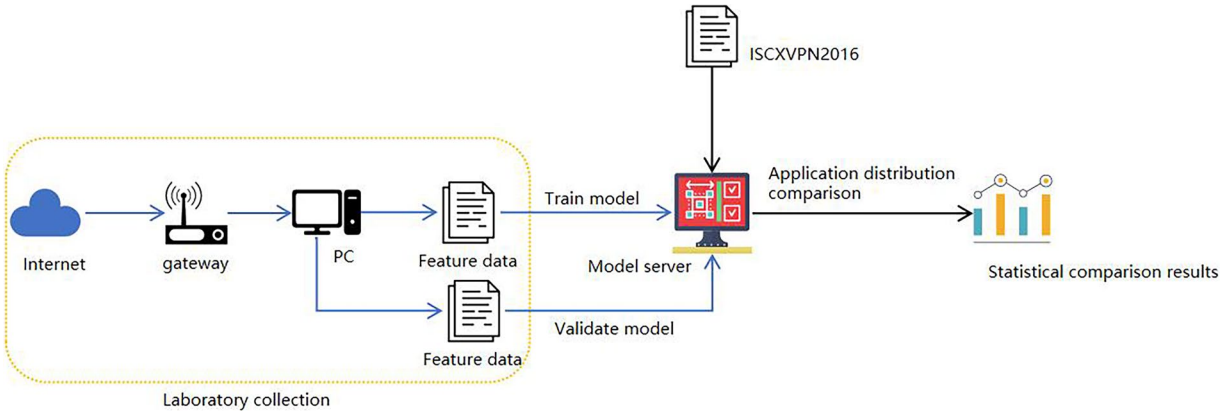


Fig. 6 Technical validation framework.

- 1. Data quality: Both datasets cover the same seven major application categories as shown in Table 3, indicating a reasonable coverage and completeness of our data. A further comparison of PCAP files' size and session counts per category (Table 6) reveals that our dataset's[12] file sizes and session numbers are similar to ISCXVPN2016, suggesting high data quality.
- 2. Data integrity: We assess the integrity by examining PCAP files' transport layer protocols, dividing them into TCP and UDP streams. We compare TCP stream completeness (counting as 1 if all three-way handshake connections in a stream are complete, and 0 otherwise) with ISCXVPN2016. Since UDP is unreliable, we compare stream-based packet loss rates (counting as 1 if there's a packet loss in a stream, and 0 otherwise). As shown in Table 7, the two datasets have results as following:

- TCP stream integrity above 99%, with our dataset[12] having a higher proportion, indicating better TCP stream integrity.
- UDP packet loss rates below 0.05%, with our dataset[12] having a lower rate, suggesting better UDP stream integrity.

In conclusion, our data integrity is strong.

Dataset	TCP Stream			UDP Stream		
	Total	Integrity	Rate	Total	Loss	Rate
Our Datasets[12]	210774	209635	99.46%	114848	31	0.03%
ISCXVPN2016[23]	365901	362644	99.11%	199373	81	0.04%

Table 7. Data integrity statistical distribution comparison.

- 3. Data authenticity: From the collected traffic from the total 45,803 users (including 19,417 BRAS users and 26,386 ONU users), 10 users were selected for an interview survey, indicating that the users' behavior reflected in these data is consistent with real-world network usage patterns.

In summary, our dataset[12] is of high quality, integrity, and authenticity, making it suitable for future relevant research purposes.

Usage Notes

To ensure optimal user experience while maintaining efficient and stable quality of service (QoS), the Network Management System (NMS) plays an indispensable role in traffic monitoring. This process is explicitly stipulated in the Service Level Agreement (SLA) governing user services.

The SLA comprehensively discloses critical details regarding traffic monitoring, including monitoring purposes, methodologies, and technical implementations, thereby ensuring full transparency for users regarding network surveillance during service utilization. This disclosure constitutes an essential element in safeguarding user rights and establishing trustworthy user-provider relationships. Furthermore, as user traffic data remains dissociated from personally identifiable information (PII) such as telephone numbers, no ethical concerns arise from this monitoring framework.

Moving forward, novel traffic data will be periodically incorporated to supersede obsolete records, contingent upon maintaining relative stability in total data volume. This dynamic updating mechanism ensures sustained dataset relevance and utility. Through this protocol, researchers may access contemporaneous network traffic data to fulfill evolving analytical requirements in telecommunications studies.

Code availability

Python codes for dataset can be obtained freely at https://github.com/zqxfree/Real-network-datasets-for-application-analysis, with no restrictions to access.

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Author contributions

Wei Jiang and Bin Zhang proposes data collection frameworks. Qixun Zhu, Conghui Liao, and Wenyong Wang conducted the validaiton experiments and analyzed the results. All authors wrote and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

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